

Desigualdad maximal doob

Teorema 1 (Desigualdad maximal de Doob). *Sea $(X_n)_n$ una submartingala respecto a la filtración $(\mathcal{F}_n)_n$ y sean $\lambda > 0$ y $n \in \mathbb{N}$, entonces*

$$(1) \quad \lambda \cdot \mathbb{P} \left(\left\{ \omega \in \Omega : \max_{1 \leq j \leq n} X_j(\omega) \geq \lambda \right\} \right) \leq \mathbb{E} \left[X_n \cdot \mathbb{1}_{\{\omega \in \Omega : \max_{1 \leq j \leq n} X_j(\omega) \geq \lambda\}} \right].$$

$$(2) \quad \lambda \cdot \mathbb{P} \left(\left\{ \omega \in \Omega : \min_{1 \leq j \leq n} X_j(\omega) \leq -\lambda \right\} \right) \leq \mathbb{E} [X_n^+] - \mathbb{E} [X_1].$$

Demostración: No se va a demostrar. ■

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